Cheng-Fu Yang

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EDUCATION

University of California, Los Angeles Sep. 2022 – Present

Ph.D. in Computer Science; Advisor: *Kai-Wei Chang*

University of California, Los Angeles Sep. 2022 – Mar. 2025

M.S. in Computer Science

National Taiwan University Sep. 2016 – Jun. 2020

B.S. in Electrical Engineering

RESEARCH INTERESTS

Multimodal agents for grounded perception, reasoning, and interaction.

Representation learning for perception, planning, and embodied intelligence.

RESEARCH EXPERIENCE

Research Scientist Intern Jun. 2026 – Sep. 2026
Topic: Multimodal post-training / next-generation SAM
Meta, FAIR Communication & Language; Mentors: *Daniel Bolya, Chay Ryali*

Research Scientist Intern Jun. 2025 – Jan. 2026
Topic: Large-scale audiovisual correspondence learning / omni-modal pre-training (CVPR 2026)
Meta, FAIR Perception; Mentor: *Bernie Huang*

Applied Scientist Intern Jun. 2024 – Sep. 2024
Topic: Precedent-based methods for customizable multimodal guardrails (COLM 2025)
Amazon AGI; Mentor: *Weitong Ruan*

Applied Scientist Intern Jun. 2023 – Sep. 2023
Topic: Diffusion policies for embodied agents (NeurIPS OWA Workshop, 2024)
Amazon; Mentor: *Feng Gao*

Research Intern Apr. 2022 – Jun. 2023
Topic: Representation learning for instruction-following agents (EMNLP 2023)
Microsoft Research; Mentor: *Yen-Chun Chen*

Research Assistant Sep. 2020 – Jun. 2022
National Taiwan University; Advisor: *Yu-Chiang Frank Wang*

Collaborative Research Project Participant Dec. 2020 – Sep. 2021
Topic: Image (Novel Object) Captioning (NeurIPS 2022)
Carnegie Mellon University; Mentor: *Yao-Hung Hubert Tsai*; Advisors: *Ruslan Salakhutdinov*, *Louis-Philippe Morency*

PUBLICATIONS

Yu Zhou*, **Cheng-Fu Yang***, Rui Sun*, Di Wu*, Sibo Peng, Mingyang Zhang,
Zhen Yang, Ruiyi Zhang, Nanyun Peng, Zhe Gan, Kai-Wei Chang. Under Submission
"OpenVLHarness: Real World Visual Agentic Tool-Use with Multimodal Memory"

Po-Nien Kung, Zhen Yang, Jeffrey Luo, **Cheng-Fu Yang**, Haikang Deng, Zi-Yi Dou,
Yinfei Yang, Nanyun Peng, Zhe Gan, Kai-Wei Chang. ICML 2026 (Spotlight)
"Learning Structured Reasoning via Tractable Trajectory Control"

Apoorv Vyas*, Heng-Jui Chang*, **Cheng-Fu Yang***, Po-Yao Huang*, Luya Gao*, Julius Richter*,
Sanyuan Chen, Matthew Le, Piotr Dollár, Christoph Feichtenhofer, Ann Lee, Wei-Ning Hsu. CVPR 2026
"Pushing the Frontier of Audiovisual Perception with Large-Scale Multimodal Correspondence Learning"

Cheng-Fu Yang, Christos Christodoulopoulos, Thanh Tran, Weitong Ruan, Rahul Gupta and Kai-Wei Chang. COLM 2025
"Customize Multi-modal RAI Guardrails with Precedent-based predictions"

Cheng-Fu Yang, Da Yin, Wenbo Hu, Heng Ji, Nanyun Peng, Bolei Zhou and Kai-Wei Chang. ICCV 2025
"Verbalize Representation Learning For Interpretable Few-shot Generalization"

Cheng-Fu Yang, Haoyang Xu, Te-Lin Wu, Xiaofeng Gao, Kai-Wei Chang, and Feng Gao. NeurIPS OWA Workshop, 2024
"Planning as Inpainting: A Generative Framework for Realistic Embodied Path Planning"

Silin Meng, Yiwei Wang, **Cheng-Fu Yang**, Nanyun Peng and Kai-Wei Chang. EMNLP, 2024
"LLM-A*: Large Language Model Enhanced Incremental Heuristic Search on Path Planning"

Zi-Yi Dou, **Cheng-Fu Yang**, Xueqing Wu, Kai-Wei Chang and Nanyun Peng. EMNLP, 2024
"Reflection-Reinforced Self-Training for Language Agents"

Cheng-Fu Yang, Yen-Chun Chen, Jianwei Yang, Xiyang Dai, Lu Yuan,
Yu-Chiang Frank Wang, and Kai-Wei Chang. EMNLP 2023
"LACMA: Language-Aligning Contrastive Learning with Meta-Actions for Embodied Instruction Following"

Wan-Cyuan Fan, **Cheng-Fu Yang**, Chiao-An Yang and Yu-Chiang Frank Wang. AAAI 2023
"Target-free Text-guided Image Manipulation"

Cheng-Fu Yang, Yao-Hung Hubert Tsai, Wan-Cyuan Fan, Ruslan Salakhutdinov,
Louis-Philippe Morency, and Yu-Chiang Frank Wang. NeurIPS 2022
"Paraphrasing is all you need for novel object captioning"

Chih-Chun Yang, **Cheng-Fu Yang**, Wan-Cyuan Fan, and Yu-Chiang Frank Wang. AAAI 2022
"Cross-Modal Mutual Learning for Audio-Visual Speech Recognition and Manipulation"

Chiao-An Yang, Cheng-Yo Tan, Wan-Cyuan Fan, **Cheng-Fu Yang**, Meng-Lin Wu, and Yu-Chiang Frank Wang. CVPR 2022
"Scene graph expansion for semantics-guided image outpainting"

Cheng-Fu Yang*, Wan-Cyuan Fan*, Fu-En Yang, and Yu-Chiang Frank Wang. CVPR 2021
"Layouttransformer: Scene layout generation with conceptual and spatial diversity"

* Equal contribution.

Awards and Honors

First Prize, Institute for Information Industry Enterprise Award at Meichu Hackathon Sep. 2019
Dean's List, National Taiwan University Sep. 2019
Dean's List, National Taiwan University Jan. 2017

Teaching Experience & Professional Service

Teaching Assistant, UCLA Fall 2025
Introduction to Machine Learning; Instructor: Kai-Wei Chang

Teaching Assistant, UCLA Fall 2024
Advanced Natural Language Processing; Instructor: Violet Peng

Teaching Assistant, UCLA Fall 2023
Introduction to Computer Science; Instructor: David Smallberg

Teaching Assistant, National Taiwan University Fall 2021
Deep Learning for Computer Vision; Instructor: Yu-Chiang Frank Wang

Teaching Assistant, National Taiwan University Fall 2019 & Spring 2020
Machine Learning; Instructor: Hung-Yi Lee

Conference Reviewer
CVPR, ECCV, ICCV, NeurIPS, ACL, EMNLP, ICLR